

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 4 MODULE, AUTUMN SEMESTER 2015-2016

COMPUTER VISION

Time allowed TWO HOURS

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer THREE Questions

Each question carries 20 marks

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn your examination paper over until instructed to do so

ADDITIONAL MATERIAL: **None**

INFORMATION FOR INVIGILATORS: **None**

1. Segmentation and Motion Detection

Gaussian mixture models are used in a variety of ways within Computer Vision. A Gaussian mixture model is defined by:

$$p(x) = \sum_i^g \pi_i \cdot p(x|\theta_i)$$

(a) In the context of image segmentation, what do g , $p(x)$, π_i and θ_i represent in this equation?

(4 marks)

(b) What is *Bayes' Rule*, and how is it used alongside Gaussian mixture models to achieve model-based image segmentation in the Expectation-Maximisation (EM) algorithm?

(8 marks)

(c) Explain how Gaussian Mixture Models can be used to detect motion in an image sequence, and give an example of a situation in which you would use a Gaussian mixture model instead of a single Gaussian.

(4 marks)

(d) Briefly describe the principle and operation of Stauffer and Grimson's adaptive motion detection scheme.

(4 marks)

2. Tracking

(a) In the context of CONDENSATION tracking what is meant by the term *particle* (2 Marks)

(b) Explain the effect on

(i) tracking performance

(ii) processing speed

of increasing the number of particles used by an implementation of CONDENSATION. (4 marks)

(b) How might you build upon CONDENSATION to successfully track the table tennis ball being bounced on the bat by the player in the image below?



(4 marks)

(c) The MCMC algorithm is better suited to tracking multiple targets than CONDENSATION, particularly when those targets might have similar appearance. Explain the difficulties each algorithm faces when tracking multiple similar targets and explain why MCMC is the better choice.

(10 marks)

3. Multi-View Stereo

- (a) What are the advantages and disadvantages associated with using a correspondence-based method to recover 3D descriptions from more than two views? (5 marks)
- (b) Briefly describe the principle underlying Baumgart's shape from silhouettes approach to 3D reconstruction. How is the method implemented within a voxel colouring framework? (4 marks)
- (c) Any voxel colouring method requires a solution to the visibility problem. Explain what this is, and how the Seitz and Dyer Voxel Colouring algorithm and Space Carving solve it. (6 marks)
- (d) A plant is placed on a computer-controlled turntable and imaged by a camera held in the jaws of a robot arm, as shown below. What problems would you face when attempting to use a volumetric approach to obtain a 3D model of the plant, and how would you address them?



(5 marks)

4. Features and Object Detection

(a) Image patches are used in the recognition and tracking of a wide variety of objects, and need to be described in ways that capture their key properties. Describe patch representations that might be used to capture the

(i) Texture

(ii) Shape

information contained in an image patch.

(10 marks)

(b) Patch descriptions often form the feature vectors used by classification algorithms.

(i) Provide the parametric form of a linear classifier

(3 marks)

(ii) Explain in simple words which criterion the SVM classifier optimises when finding its optimal parameter set.

(3 marks)

(iii) You have trained a classifier to recognise a particular class of objects. How would you evaluate your method?

(4 marks)